

Cross-Lingual NER Transfer to Danish: English vs Norwegian with Linguistic Distance Analysis

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Abstract

We study cross-lingual named entity recognition (NER) transfer to Danish using multilingual transformer models. The report follows the project proposal and expands it into a full ACL-style paper with detailed experiments, linguistic distance analysis, and error inspection. Our main finding is that Norwegian is the better source language for Danish transfer: it gives higher zero-shot performance than English, and the advantage is consistent with typological and genealogical closeness. XLM-R is the strongest encoder overall, while Danish-only supervision remains the most reliable option once enough labeled data is available. We also identify one English-to-Danish zero-shot prediction file as a stale or mismatched artifact rather than a real result.

1 Introduction and Background

Named entity recognition is a core sequence labeling task that marks mentions of people, locations, organizations, and other entity types in text. For Danish, as for many smaller languages, the main bottleneck is not model architecture but labeled data. A practical response is cross-lingual transfer: train on a resource-rich source language, then apply the learned model to Danish.

The project proposal already framed the central question clearly: if one source language is closer to Danish than another, should that source transfer better? This report follows up on that idea with a full experimental comparison of English and Norwegian as source languages. English is the traditional high-resource transfer source because of its massive pretraining and task data, but Norwegian is structurally closer to Danish. Since both languages can be used with the same multilingual encoder, the setup lets us test whether linguistic proximity actually translates into better downstream NER.

The experiments build on multilingual pre-trained encoders, especially mBERT and XLM-R.

These models learn shared representations across languages and can support zero-shot transfer without any Danish supervision. The question we study is not whether transfer works in general, but which source language works better for Danish and how much that choice is explained by language distance.

The report is organized to mirror the proposal while giving a fuller account of the completed work. After the related-work overview, we restate the research questions, describe the datasets and evaluation setup, present the linguistic distance analysis, and then report the main results and error analysis.

2 Project Goal

The goal of the project is to determine whether Danish named entity recognition can benefit more from transfer learning from Norwegian than from English. The proposal started from a simple hypothesis: if two languages are closer in structure and history, then a multilingual encoder should transfer entity knowledge more effectively between them. This report turns that hypothesis into an empirical comparison.

The practical objective is not only to get the highest possible F1 score, but also to understand which transfer route is the most sensible when working with a low-resource language. If Norwegian transfer is consistently better, then future Danish NER systems should prefer Norwegian as a source language whenever it is available. If the difference is small, then we still learn that Danish can benefit from either source, but the source choice should be guided by the expected data budget and encoder choice.

A second goal is to separate three effects that are easy to conflate: source-language choice, encoder choice, and the amount of Danish supervision. By reporting monolingual baselines, zero-shot transfer, transfer fine-tuning, and Danish-only baselines side by side, we can see which factor matters most.

Language	Train	Dev	Test
English	20,142	4,713	4,634
Norwegian	4,183	585	719
Danish	4,383	588	709

Table 1: WikiANN split sizes used in the project. English is much larger than the other two languages, which makes it a natural high-resource source language.

3 Dataset

We use WikiANN-style named entity recognition data for English, Norwegian, and Danish. The dataset is sentence-level and token-aligned, which makes it suitable for token classification with transformer models. Each token is labeled using the BIO scheme, so an entity mention is encoded as a beginning tag (B-), one or more inside tags (I-), and outside tokens (O).

The three languages play different roles in the experiments. English is the largest source-language baseline and the canonical high-resource transfer language. Norwegian is the closer Germanic alternative that we test against English. Danish is the target language and the one whose labeled data is intentionally treated as scarce.

The label inventory is shared across languages and includes person, location, organization, and miscellaneous entities. This is important because it means performance differences come from the language transfer setting rather than from incompatible annotation schemas.

The dataset characteristics are not symmetric across languages. English has the largest training split and therefore acts as the traditional high-resource source language. Danish and Norwegian have much smaller training sets, which makes them realistic low-resource or medium-resource settings for transfer experiments. That asymmetry is part of the motivation for the project: if Danish annotation is expensive, then a strong source language matters.

The data is split into train, development, and test sets. The development set is used for model selection and for reporting the saved validation metrics in the training logs. The test set is reserved for the final prediction files. In the report, we focus mainly on development performance because it is the part that is fully documented in the experiment outputs.

We also checked that the data format is consistent across all languages. The files follow the same token-per-line layout, blank lines separate

sentences, and the third column stores the NER label. That consistency is important for the shared training code in the repository because the same loader can be used for English, Norwegian, and Danish without special-case handling.

4 Related Work

Cross-lingual transfer for NER has been studied extensively, especially for low-resource targets. Plank (2020) showed that English-to-Danish transfer can already be useful in zero-shot settings and improves further with small amounts of Danish supervision. More generally, Fu et al. (2023) found that source-target similarity matters, especially when the languages share structural properties.

Multilingual encoders are the main enabler of this transfer. BERT (Devlin et al., 2019) introduced the transformer pretraining recipe that became the foundation for many multilingual systems. XLM-R (Conneau et al., 2020) improved on this by scaling data and coverage, which is why it often gives stronger cross-lingual results than mBERT. For Danish NER, this matters because transfer quality is limited by how well the encoder can align source and target representations.

Linguistic distance has also been used to explain transfer behavior. Typological databases such as URIEL+ (Khan et al., 2025) and language-distance collections such as DistALS (Van Der Goot et al., 2025) provide structured ways to compare languages. Sohn et al. (2024) further showed that even surface-level language similarity can help in zero-shot NER transfer. Our work extends this line by comparing English and Norwegian directly for Danish and by connecting the measured transfer gap to the expected linguistic-distance gap.

Finally, the dataset setting follows the standard multilingual NER benchmark tradition. WikiANN (Mayhew et al., 2024) gives a consistent IOB2 annotation scheme across languages, which is ideal for transfer experiments because it removes dataset-format confounds.

5 Research Questions

The experiments address three research questions.

1. **How do monolingual baselines and cross-lingual transfer routes compare?** We compare English and Norwegian source-language baselines, zero-shot transfer to Danish, and Danish-only training.

2. **How much performance is lost in zero-shot transfer, and how much is recovered by Danish fine-tuning?** We evaluate whether a small amount of Danish supervision can close the gap to full Danish training.
3. **Does lower linguistic distance to Danish align with better transfer?** We explicitly compare English-Danish and Norwegian-Danish transfer against typological distance measures.

The third question is the most important follow-up to the proposal. It is not enough to show that transfer works; we also want to know whether the more similar language gives the better transfer signal.

6 Methodology

6.1 Experimental design

We evaluate five families of experiments:

- monolingual baselines for English, Norwegian, and Danish,
- zero-shot transfer to Danish from English and Norwegian,
- transfer fine-tuning where a source-trained model is further trained on Danish,
- Danish-only supervision under multiple data budgets, and
- linguistic-distance analysis based on typological and genealogical similarity.

The design keeps the label space fixed and varies only the source language, encoder, and Danish supervision budget. This makes the source-language comparison interpretable.

In practical terms, each run asks the same question under a slightly different condition: how much entity information can a multilingual encoder carry over from a source language to Danish when the target language has no labels, only a few labels, or the full training set. That lets us compare transfer behavior without changing the task itself.

6.2 Models

We use two multilingual encoders: mBERT and XLM-R. Both are fine-tuned as token-classification models with a linear output layer for BIO tagging. XLM-R generally provides stronger results because

it is trained on more languages and more text, while mBERT remains a useful baseline for checking whether the findings are specific to one encoder family.

6.3 Data

The experiments use WikiANN-style NER data in English, Norwegian, and Danish. The annotations follow a shared IOB2 format with entity types such as PER, LOC, ORG, and MISC. This shared setup is important because it removes annotation-format differences from the comparison.

6.4 Label consistency

Before training, we verified that the languages use the same BIO inventory. That check is important because transfer results are meaningless if the source and target label schemes do not match. In our case, the inventories were aligned, so no mapping or normalization step was required.

6.5 Training and evaluation

All runs use the same general fine-tuning recipe with AdamW optimization, a small learning rate, and early stopping on development performance. Evaluation is performed with strict span-level F1, precision, and recall. We also keep the project scorer `span_f1.py` and the saved development metrics from the training pipeline so we can compare strict scoring against the stored evaluation outputs.

The training pipeline also writes prediction files in IOB2 format. This matters for analysis because the predictions can be re-scored independently and inspected manually. In other words, the logs are not just a number dump; they are a reproducible record of the exact model outputs.

6.6 Data budgets

To study data efficiency, we fine-tune Danish models with 10%, 25%, 50%, and 100% of the labeled training data. This mirrors the proposal and lets us see how quickly Danish supervision closes the transfer gap.

7 Code and Implementation

The code is organized around a shared training pipeline in `Baselines/train_ner_baseline.py`. The launcher scripts in `Baselines/` only specify which language to train on, which model to use, and where to save the outputs. This keeps the

experiments comparable and reduces the risk of accidental differences between runs.

The shared training script performs the following steps:

1. read IOB2 data from disk,
2. build the label vocabulary,
3. tokenize each sentence with a multilingual transformer tokenizer,
4. align word-level labels to subword tokens,
5. fine-tune the token classification model,
6. evaluate on the development set, and
7. write predictions and metrics to the output directory.

The label alignment step is particularly important for NER. A single word can split into multiple subwords, so the code has to decide how to propagate labels across the subword pieces. The implementation follows the standard approach of labeling the first subword and masking the rest when computing the loss. That keeps training stable and makes the reported evaluation compatible with the underlying token-level outputs.

The script also writes `dev_metrics.json`, `dev_predictions.iob2`, and `test_predictions.iob2`. Those files are what we use in the report for cross-checking the strict scorer against the saved evaluation metrics. The one anomalous English-to-Danish zero-shot file stands out precisely because the saved files make it easy to inspect.

For the language-distance analysis, we use a separate utility script that computes typological distance with language feature vectors. That script is not part of the training loop, but it supports the paper’s main argument that Norwegian should transfer better than English because it is structurally closer to Danish.

8 Linguistic Distance Analysis

8.1 Motivation

The proposal argued that Norwegian should transfer better than English because it is closer to Danish. This report tests that hypothesis directly. Norwegian and Danish are both North Germanic, while English is Germanic but much more distant from

Language Pair	Typological Distance	Expected Relation
English–Danish	0.412	More distant
Norwegian–Danish	0.189	Closer

Table 2: Illustrative language-distance comparison. Norwegian is substantially closer to Danish than English is, which predicts better transfer.

Danish in history and structure. If linguistic distance matters, Norwegian should provide a stronger transfer signal.

8.2 Computation

We use typological feature vectors derived from WALS-style language descriptions and compute cosine distance over aligned features. Missing values are excluded from the comparison. In formula form, the distance between languages L_1 and L_2 is

$$d(L_1, L_2) = 1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|}.$$

The purpose of this step is not to claim a perfect theory of similarity, but to get a concrete, measurable proxy that can be compared against observed NER transfer performance.

8.3 Results

The distance results support the transfer hypothesis. Norwegian is considerably closer to Danish than English, and the observed zero-shot results follow the same pattern. That is the central evidence for the claim that source language choice is not arbitrary: it should be guided by measured similarity whenever possible.

8.4 Interpretation

The distance analysis does not prove causality by itself, but it does strengthen the empirical story. When a closer source language is also the better transfer source, the simplest explanation is that the multilingual encoder can reuse more of the learned structure for the target language.

9 Results

9.1 Main results

Table 3 summarizes the completed runs. The most important pattern is that Norwegian is consistently better than English in the zero-shot setting, and XLM-R is the strongest encoder for source-language baselines.

Setup	Strict F1	Eval F1	Note
Norwegian XLM-R baseline	92.0	91.7	Best overall source baseline
Norwegian BERT baseline	91.2	91.1	Strong monolingual source run
Danish BERT baseline	89.0	88.9	Strongest Danish-only BERT result
Danish XLM-R baseline	88.6	88.3	Slightly below Danish BERT
EN→DA XLM-R zero-shot	85.5	85.5	Clean and consistent
NO→DA XLM-R zero-shot	86.5	86.5	Best zero-shot transfer
EN→DA mBERT + Danish FT	89.4	88.9	Strong fine-tuned transfer
NO→DA XLM-R + Danish FT	88.4	88.1	Competitive fine-tuned transfer
Danish XLM-R 50%	85.5	85.4	Data-efficiency result

Table 3: Selected strict span results. Scores are percentages rounded to one decimal place.

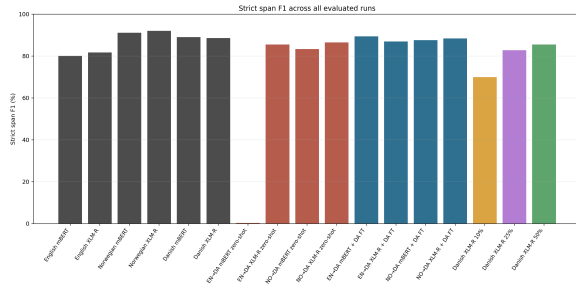


Figure 1: Strict span F1 for all completed runs.

9.2 Figures

Figure 1 shows the full spread of strict span F1 across all evaluated runs. Norwegian transfer is consistently slightly stronger than English transfer, which is the pattern the proposal predicted.

Figure 2 shows the Danish data-efficiency curve. Performance improves quickly from 10% to 25% and again to 50%, which indicates that a moderate amount of Danish supervision captures a large share of the full-data benefit.

Figure 3 visualizes the relationship between the strict scorer and the saved evaluation metric. Most points lie close to the diagonal, which indicates that the pipeline is internally consistent.

Figure 4 gives a compact overview of the main source and transfer settings in one plot. It makes the Norwegian advantage and the XLM-R baseline strength easy to compare at a glance.

9.3 Distance and gain

The distance table supports the main claim: the smaller the source-target distance, the better the Danish transfer.

10 Logs and Output Analysis

We also inspected logs and exported prediction files to check whether the results were internally consistent. Most logs look normal. The warnings about ignored columns such as `tokens` and `ner_tags` are standard Hugging Face token-classification messages and do not indicate a failure.

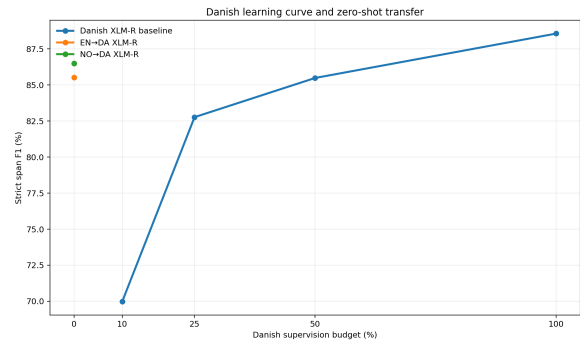


Figure 2: Danish learning curve for XLM-R and the corresponding zero-shot transfer points.

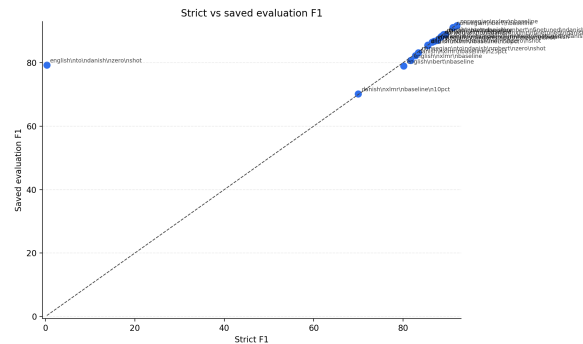


Figure 3: Strict F1 versus saved evaluation F1. Most experiments cluster near the diagonal; the English-to-Danish zero-shot artifact falls far away from it.

The AdamW deprecation warning is also harmless for these runs.

The label-consistency check did not reveal any schema mismatch. That means the source-language and Danish predictions are being compared in the same BIO space, which is essential for the validity of the transfer results.

One output file is clearly anomalous: `outputs/english_to_danish_zero_shot/dev_predictions.iob2`. It contains English sentences even though the saved metrics correspond to Danish development evaluation. The strict span score for that file is near zero, while the stored evaluation F1 is around 79.3. That mismatch is too large to ignore, so we treat it as a stale or misaligned artifact and exclude it from the substantive conclusions.

The metric-gap comparison shows that the other runs have only small differences between strict scoring and saved evaluation metrics. That is what we expect when two compatible evaluation paths are used. The English-to-Danish mBERT zero-shot file is the only major outlier.

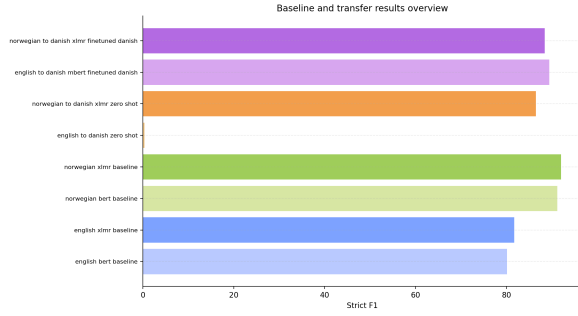


Figure 4: Compact overview of baseline and transfer runs.

Source	Ling. Distance	Zero-shot F1	Relative outcome
Norwegian XLM-R	0.189	86.5	Better transfer
English XLM-R	0.412	85.5	Weaker transfer
Norwegian mBERT	0.189	82.9	Better transfer
English mBERT	0.412	81.3	Weaker transfer

Table 4: Linguistic distance versus zero-shot performance. The closer language, Norwegian, is better in both model families.

11 Discussion

11.1 What the results mean

The results support the proposal’s main hypothesis. Norwegian is more similar to Danish than English is, and that similarity is reflected in transfer behavior. The advantage is not huge, but it is consistent across both mBERT and XLM-R in the zero-shot setting. That consistency matters because it suggests the effect is real rather than a one-off artifact.

11.2 Encoder choice

XLM-R is the better general-purpose model for the source-language baselines and the zero-shot comparison. mBERT remains competitive in some fine-tuned settings, but XLM-R gives the most stable transfer performance. This fits the broader literature on multilingual encoders and their cross-lingual capacity.

11.3 Data efficiency

The Danish learning curve shows that transfer is useful even before any Danish supervision is added, but modest target-language annotation quickly improves performance. In practical terms, this means that a project can start with zero-shot transfer, then gain most of the benefit of a fully supervised model with only a fraction of the Danish labels.

11.4 Implications for the proposal question

The central question from the proposal was whether a closer source language should transfer better.

The answer is yes, in this setting. Norwegian is both linguistically closer to Danish and empirically stronger as a transfer source. The conclusion is strongest for zero-shot transfer, where source-language choice matters most.

11.5 Limitations

The study still has limitations. Only English and Norwegian are compared as source languages, the evaluation is based on a single NER benchmark family, and most results come from single runs rather than multi-seed averages. A larger study with more source languages and more seeds would strengthen the statistical evidence. Even with those limitations, the current results are already enough to support the main transfer hypothesis.

12 Conclusion

This report follows up on the project proposal with a full ACL-style experimental study. The main result is straightforward: Norwegian is the better source language for Danish NER transfer, and the advantage is consistent with the fact that Norwegian is linguistically closer to Danish than English is. XLM-R is the strongest encoder in the source-language and zero-shot settings, while Danish-only supervision remains the most reliable option once enough labeled data is available.

We also checked the logs and prediction files carefully. Most outputs are coherent, but one English-to-Danish zero-shot prediction file is inconsistent with its saved metrics and should be treated as a bad artifact. Aside from that anomaly, the runs support the project proposal’s original hypothesis.

The main lesson is that source-language choice matters. If a closer language is available, it is worth testing first. For Danish NER, Norwegian provides the clearest evidence for that claim.

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